CommonsLens: Progress Report (M2)

TEAM-89

Hackathon Problem Statement 3

1. Progress Summary

The M1 pipeline runs end to end on real Wikimedia Commons data: crawl through the MediaWiki API → clean categories → temporal split → frozen CLIP ViT-B/32 encoder → four scorers → calibrated fusion → hierarchy decoding → explained suggestions in a web demo. We ran three successively larger experiments (up to 6,303 files and 24,486 categories), fixed a validation flaw they exposed, and tested the demo on our own photos, which led to four usability fixes. Two M1 components are still pending: graph-smoothed prototypes and conformal sets.

2. Changes Since M1

M1 component	Status
Frozen encoder (SigLIP/CLIP)	Done: CLIP ViT-B/32; SigLIP supported, comparison pending
kNN retrieval + neighbour explanations	Done; K, τ tuned on validation
Text prototypes	Done (prompt ensembles); multilingual labels pending
Supervised head (ASL, text-initialised)	Done; threshold lowered to $\geq$3 examples (466 categories)
Metadata scorer	Done: title/description embedding, word overlap, year agreement; no geotag yet
Graph-smoothed prototypes	**Pending** (next task)
Calibrated LR fusion on candidate union	Done
Hierarchy-aware decoding	Done; strength now tuned on validation
Conformal sets / abstention	**Partial**: calibrated "Unsure" abstention; conformal sets pending
Temporal split, dedup, uploader grouping	Done (pHash + cosine $\geq$0.95; uploader-disjoint option)
Robustness	Done: low-quality flags; fusion trained on blurred, compressed and downscaled copies
Continual learning	Done: index append, live new categories, rejects as negative votes

Added beyond M1. (1) *Realistic validation:* the vocabulary is built only from data that precedes the validation period; before, every validation label was "known" and fusion was tuned on an easy set. (2) *Content type first:* each result starts with Object / Scene / Event / Abstract concept, blending a vote of the suggested categories with a zero-shot check. (3) *Bias audit:* region-balanced seeds (8 countries per continent), optional uniformly random files, accuracy per region and topic. (4) *Demo fixes from our own testing* (Section 5). (5) *Tester problem reports* stored with what the system showed, for an open feedback loop.

3. Experiments

Labels are editor-assigned categories after removing hidden and maintenance ones; splits are by upload time. **Run 1**: 1,279 files, 107 topical seeds. **Run 2**: 3,416 files, more per category. **Run 3**: 6,303 files (4,429 / 937 / 937) from 155 seeds plus 1,500 random files, with realistic validation; 24,486 categories; K=50, τ=0.1.

Test, image + metadata	Run 1	Run 2	Run 3
Test images	190	510	937
P@1	.284	.339	**.350**
F1@1	.148	.199	**.237**
F1@5	.119	**.160**	.153
R@10	.238	.310	**.330**
F1@5, image only	.089	**.112**	.084
Label coverage	.335	.377	.362
Val − test F1@5 gap	–	.135	**.014**

Run 3 ablation

Ranking	F1@5	R@10
Fusion	**.153**	**.330**
Lexical only	.100	.203
Head only	.070	.173
kNN only	.069	.156
Meta text only	.052	.170
Zero-shot only	.005	.020

Recall@5 by frequency

Bucket	labels	R
unseen	2,412	.004
tail 1–4	318	.214
torso 5–19	217	.434
head 20+	380	.830

Bias audit, F1@5 (Run 3)

Group	n	+meta	img
Objects	318	.192	.140
Scenes	109	.232	.162
Events	20	.233	.200
Abstract	127	.050	.029
Europe	42	.278	.037
Africa	71	.204	.009
S. America	48	.103	.009
Asia	47	.087	.038
N. America	43	.032	.012

Robustness, F1@5

Degradation	+meta	img
none	.153	.084
blur r=2	.155	.079
blur r=5	.147	.054
JPEG q10	.151	.072
down ×.15	.151	.066

Efficiency (on a Mac, no server GPU): p50 19.6 ms, p95 114.8 ms per image end to end; 0.89 ms for scoring and ranking; 5,366 indexed images and 24,486 categories in 58.3 MB.

4. Findings

- **Realistic validation fixed model selection:** the val–test gap fell from .135 to .014, and the tuned hierarchy rule (strength chosen on validation) no longer costs accuracy.
- **More data improves the top of the list:** P@1 .284→.350, R@10 .238→.330. F1@5 is flat on Run 3's harder test set, whose region-balanced seeds add place-specific categories.
- **The tail is the bottleneck**, as M1 predicted: recall@5 is .830 for categories with $\geq$20 examples but .004 for unseen ones, which hold 72% of test labels. This is exactly what graph-smoothed prototypes target.
- **Fusion** beats the best single scorer by 53%; **metadata** is essential for places (Africa .204 vs .009 image only); under heavy blur F1@5 drops 4% with metadata vs 36% image only.
- **Zero-shot name matching alone collapses** over 24k names (.005), motivating smoothed prototypes and SigLIP. Regional accuracy varies widely; groups are

small, so differences are indicative.

5. Progress on Testing

- **Evaluation harness** implements every M1 metric (P/R/F1@k, mAP, hierarchy-aware precision, frequency buckets, type, resolution, degradations, latency, memory, ablations) plus the region audit and content-type accuracy.
- **67 automated tests**, including an offline end-to-end run (train → evaluate → predict → update) and encoder API checks on tiny CLIP and SigLIP models.
- **Hands-on testing with our own photos** found and fixed: people pulled towards "Marathons" and floor tiles towards "Textiles" (centre-weighted image embedding); a cathedral typed as Object (buildings and landmarks now count as Scene); bare country names as noise (hidden); abstention applied to the wrong score (now based on content-type confidence, threshold 50%).
- **Continual learning verified:** new categories and accepted feedback are suggested immediately without retraining.

6. Plan for the Remaining Time

1. **Graph-smoothed prototypes** (M1's core idea): blend each category's text embedding with its image centroid and its parent/sibling prototypes; ablate tail recall with and without.
2. **Conformal suggestion sets:** split conformal calibration on validation; report coverage vs target, set size and abstention rate against fixed top-k.
3. **Encoder comparison** (SigLIP vs CLIP) and an **unbiased test set** of the newest Commons uploads.
4. **Reviewed score** on a sample of "wrong" suggestions; final report, slides and demo video.